From Dead Cross-Sectional Momentum to Belief-State Deep Time-Series Momentum: An Honest, Cost-Aware Study on a Diversified Asset Basket

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June 7, 2026

Abstract

We study momentum trading through the lens of honest, cost-aware, out-of-sample evaluation, and report a deliberately two-sided result. First, a *negative*: on US large-cap equities (S&P 100/500), cross-sectional momentum—ranking names against one another—is dead net of realistic costs over 2006–2024, and a Bayesian online changepoint overlay does not revive it. Second, a *positive*: the same trading logic applied as *time-series* momentum to a diversified basket of eighteen liquid exchange-traded funds (equities, bonds, gold, commodities, the US dollar, real estate, sectors) is alive. We construct a per-asset belief state with a Kalman local-linear-trend filter and Bayesian online changepoint detection, feed it together with multi-horizon trend features to a small LSTM "Deep Momentum Network" trained on a portfolio Sharpe-ratio objective net of turnover costs, and select hyperparameters by a *nested* walk-forward that never lets the validation choice touch the out-of-sample test. The learned model attains an out-of-sample (2016–2024) information ratio of 0.58 (Newey–West $t \approx 1.98$, Deflated Sharpe Ratio 0.67 over five trials), beating a fixed multi-horizon rules baseline (0.47) at lower drawdown. We isolate the source of the edge: it is *asset-class diversity*, not name count—five genuinely uncorrelated instruments (average pairwise correlation 0.03) more than double the risk-adjusted return of twenty-two sector-diversified equities (correlation 0.41). Finally we show the practical use: a return-stacked overlay (full equities plus a half-sized trend sleeve) beats buy-and-hold S&P 500 out of sample on both absolute return and Sharpe. All experiments are reproducible from free Yahoo data, net of a transparent cost model, with pre-registered honesty: a correctly-inferred null is reported as such.

1 Introduction

Momentum—the tendency of recent winners to keep winning—is among the most studied phenomena in empirical finance, and one of the few with out-of-sample persistence across decades and asset classes [15]. The deep-learning literature has enriched it: Deep Momentum Networks inject learned trading rules into a volatility-scaling framework and optimize a Sharpe-ratio loss directly [10]; Slow Momentum with Fast Reversion augments this with changepoint detection to survive trend reversals [16]; and the Momentum Transformer replaces recurrence with attention [17]. A recurring weakness of the broader reinforcement- and deep-learning trading literature is evaluation: raw price windows as state, in-sample tuning, and performance reported without rigorous walk-forward or cost accounting [4].

This paper takes the honest-evaluation stance seriously and lets it dictate the result rather than the other way around. We make three contributions, organized as a single empirical narrative.

(i) A motivating negative. We first show that the obvious adaptation—cross-sectional momentum on US large-cap equities, dollar-neutral long-winners/short-losers—is *dead* net of costs in 2006–2024, and that a Bayesian online changepoint overlay does not save it (Section 3). This is consistent with the decay of equity momentum in liquid large-caps and is reported plainly, not buried.

(ii) A belief-state Deep Momentum Network that works on the right universe. We construct a per-asset belief state from a Kalman local-linear-trend filter (filtered velocity, a trend t-statistic, and the standardized innovation) and Bayesian online changepoint detection [7, 1], feed it with multi-horizon volatility-normalized returns to a small shared LSTM, and train on a portfolio Sharpe objective net of turnover costs in the direct- reinforcement spirit of [14]. Crucially, hyperparameters are chosen by a *nested* walk-forward—validation-only selection with early stopping—so the out-of-sample number is honest by construction (Section 4). On a diversified eighteen-ETF basket the learned model beats a strong fixed-rule baseline out of sample (Section 5).

(iii) Where the edge comes from, and how to use it. We isolate the mechanism—asset-class diversity, quantified by average pairwise correlation—and show it dominates name count (Section 6); and we demonstrate a practical deployment in which a return-stacked trend overlay beats buy-and-hold equity out of sample (Section 7).

We are deliberately conservative about magnitudes. Better state estimation extracts a weak signal more cleanly; it does not manufacture signal. Under the Fundamental Law of Active Management [3], our levers are signal quality and breadth of *independent* bets, not raw instrument count. The realistic target is a diversified information ratio of order 0.4–0.7, and every number below is reported net of a transparent cost model.

2 Related Work

Momentum and deep momentum. Time-series momentum [15] trades each asset on its own past return, volatility-scaled; it is the economic baseline our model must beat. Deep Momentum Networks [10] learn position sizing end-to-end under a Sharpe loss; the Slow Momentum / Fast Reversion architecture [16] adds changepoint detection to handle turning points; the Momentum Transformer [17] is the attention-based successor. Our model is a deliberately small LSTM Deep Momentum Network whose novelty is not capacity but the *belief-state* inputs (a single Kalman filter supplies both the trend and the changepoint signal) and the *nested*, cost-aware evaluation.

Filtering and changepoints. The Kalman filter [7] and structural time-series models give the linear-Gaussian belief; Bayesian online changepoint detection [1] supplies a per-asset regime-break severity. We note that simple linear models are competitive with elaborate sequence models at this frequency and signal-to-noise [18], which motivates a small network plus an informative inductive bias rather than raw capacity.

Honest evaluation. Multiple-testing inflation of backtested Sharpe ratios is corrected by the Deflated Sharpe Ratio [2]; walk-forward discipline, purged cross-validation, and the probability of backtest overfitting are developed in [12]. We use Newey–West t-statistics throughout and report the Deflated Sharpe Ratio against the number of configurations tried.

3 A Motivating Negative: Cross-Sectional Equity Momentum

We first test the natural equity adaptation: each day, rank the investable universe by a 12−1 momentum score, go long the top decile and short the bottom decile, dollar-neutral and volatility-scaled, rebalanced and held, net of a 5 bps spread on turnover. On the S&P 100 (2000–2024) and on a 414-name S&P 500 sample (2006–2024), every variant we tried lost money out of sample: plain cross-sectional momentum, a Kalman-velocity ranking, and a significance-gated belief variant all returned information ratios between −0.16 and −1.34 on the S&P 100 and between −0.28 and −0.47 on the broader sample across daily to quarterly rebalancing. A market-level Bayesian changepoint overlay did not help (it moved the information ratio from −0.36 to −0.37): the losses are not concentrated in detectable regime breaks but bleed across a structurally low-momentum, highly-correlated, efficient large-cap universe. We report this as a clean null and move to the setting where the same logic is alive (Figure 1).

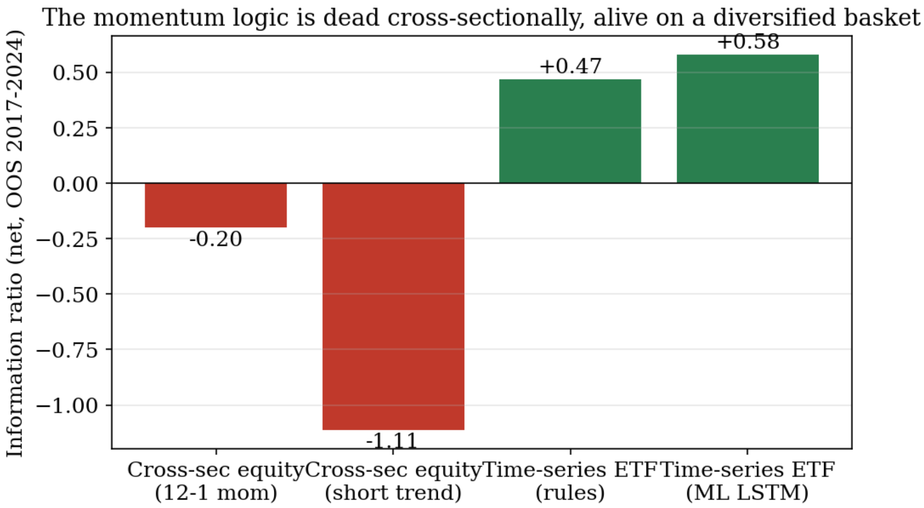

Figure 1: Out-of-sample (2017–2024), net-of-cost information ratio. Cross-sectional momentum on US large-cap equities (realistic spread plus borrow) is negative; the same trend logic applied as time-series momentum to a diversified ETF basket is positive, and the learned Deep Momentum Network improves on the fixed rule.

4 Belief-State Features and the Deep Momentum Network

4.1 Universe and feature vector

Universe. A fixed basket of eighteen liquid ETFs spanning asset classes: US large/tech/small-cap, developed and emerging international equity; long, intermediate, investment-grade and high-yield bonds; gold, silver, broad commodities and oil; the US dollar; real estate; and three equity sectors. Daily adjusted prices from Yahoo Finance, 2007–2024.

Features. For each asset j and day t we form an $F = 10$ feature vector: five volatility-normalized cumulative returns over horizons $\{1, 21, 63, 126, 252\}$ days, the log realized volatility, and four *belief-state* signals produced by the two filters below — the Kalman trend velocity $\hat{v}_t$, its significance $\hat{v}_t/\sqrt{P_t^{vv}}$, the standardized innovation $\nu_t/\sqrt{S_t}$, and the changepoint severity $s_t \in [0, 1]$. The first six are standard momentum/scale descriptors; the distinctive part of the input is the belief block, which encodes not just *how* the price moved but the filter's *confidence* in the trend and its assessment of *regime stability*. We detail the two filters next, as they are central to the method and to the paper's contribution.

4.2 Local-linear-trend belief state

We model the log price y_t of each asset as a Gaussian linear state-space system with latent level μ_t and velocity v_t, collected in $\alpha_t = (\mu_t, v_t)^\top$:

$$\alpha_t = T\alpha_{t-1} + \eta_t, \qquad y_t = Z\alpha_t + \varepsilon_t, \qquad T = \left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right), \quad Z = (1 \;\; 0), \tag{1}$$

with $\eta_t \sim \mathcal{N}(0, Q)$, $Q = \operatorname{diag}(q_\mu, q_v)$, and $\varepsilon_t \sim \mathcal{N}(0, r)$. This is Harvey's local-linear-trend model [5]: the level integrates the velocity and the velocity follows a random walk, so the filter tracks a smoothly time-varying drift. Given the filtered posterior $\alpha_{t-1} \mid y_{1:t-1} \sim \mathcal{N}(a_{t-1}, P_{t-1})$, the Kalman recursion [7] alternates a one-step prediction

$$a_{t|t-1} = Ta_{t-1}, \qquad P_{t|t-1} = TP_{t-1}T^\top + Q, \tag{2}$$

and an update driven by the innovation $\nu_t = y_t - Za_{t|t-1}$ with variance $S_t = ZP_{t|t-1}Z^\top + r$:

$$K_t = P_{t|t-1}Z^\top S_t^{-1}, \qquad a_t = a_{t|t-1} + K_t\nu_t, \qquad P_t = (I - K_tZ)\,P_{t|t-1}. \tag{3}$$

Under this Gaussian model the recursion returns the exact filtering posterior, whose mean is the minimum-variance — and, being linear, the best linear unbiased — estimator of α_t given $y_{1:t}$ [7, 5]. We fix the variances across assets at $(q_\mu, q_v, r) = (10^{-5}, 10^{-7}, 10^{-6})$ — deliberately small process noise, yielding a slowly varying trend — rather than estimating them per asset, for parsimony and to avoid per-asset overfitting.

From the posterior we read three signals. The **velocity** $\hat{v}_t = (a_t)_2$ is the filtered drift. Its **significance** $\hat{v}_t/\sqrt{P_t^{vv}}$, with $P_t^{vv} = (P_t)_{22}$, is a local t-statistic: it is large only when the trend is both fast *and* precisely estimated, so it discounts trends the filter is unsure of. The **standardized innovation** $\nu_t/\sqrt{S_t}$ is the one-step-ahead surprise; under correct specification it is i.i.d. $\mathcal{N}(0,1)$, so persistent large values flag model misfit — precisely the regime stress the changepoint detector formalizes next.

4.3 Online changepoint detection

The Kalman significance fades smoothly; abrupt regime breaks call for a complementary detector. We run the Bayesian online changepoint detector of Adams and MacKay [1] on each asset's returns x_t. It maintains a posterior over the *run length* $r_t \in \{0, 1, \dots\}$, the number of steps since the last changepoint, under a constant hazard $H(r) = h$ (we use $h = 1/250$, one expected break per trading year). Each run carries a conjugate Normal–Normal predictive: for a run of length r with n_r observations summing to Σ_r and fixed observation variance σ^2, the run mean has posterior precision $\tau_r = \kappa_0^{-1} + n_r/\sigma^2$ and mean $m_r = (\mu_0\kappa_0^{-1} + \Sigma_r/\sigma^2)/\tau_r$, giving the predictive density $\pi_t^{(r)} = \mathcal{N}\big(x_t;\, m_r,\, \sigma^2 + \tau_r^{-1}\big)$. The joint run-length recursion is

$$p(r_t = r{+}1, x_{1:t}) = p(r_{t-1} = r, x_{1:t-1})\,\pi_t^{(r)}\,(1-h) \qquad \text{(growth)}, \tag{4}$$

$$p(r_t = 0, x_{1:t}) = \textstyle\sum_r p(r_{t-1} = r, x_{1:t-1})\,\pi_t^{(r)}\,h \qquad \text{(changepoint)}, \tag{5}$$

normalized to the run-length posterior $P(r_t \mid x_{1:t})$. We summarize it by the **severity**

$$s_t = \Pr(r_t < k \mid x_{1:t}) = \sum_{r=0}^{k-1} P(r_t = r \mid x_{1:t}), \qquad k = 5, \tag{6}$$

the posterior mass on short run lengths. In a stable regime the mass concentrates on large r and $s_t \approx 0$; when the data cease to fit the incumbent run the mass collapses onto small r and $s_t \to 1$ — a graded "regime just changed" signal that responds to a change in *behaviour*, not to a single adverse return.

Remark (why severity, not $P(r_t{=}0)$). A natural but mistaken detector is the normalized changepoint posterior $P(r_t{=}0 \mid x_{1:t})$. Write $M_t = \sum_r p(r_{t-1}{=}r, x_{1:t-1})\, \pi_t^{(r)}$ for the shared predictive mass. From the recursion the unnormalized changepoint term equals $h\, M_t$ and the full normalizer equals $\big[h + (1-h)\big] M_t = M_t$, so

$$P(r_t{=}0 \mid x_{1:t}) \;=\; \frac{h\, M_t}{M_t} \;=\; h \tag{7}$$

identically (up to the negligible run-length truncation), independent of the data — it carries no information. The collapse of mass onto short, but not necessarily zero, run lengths does, which is why we use s_t. We initially used $P(r_t{=}0)$ and observed exactly this degeneracy; replacing it with s_t restored a usable signal.

4.4 Architecture and training objective

A single LSTM [6] is shared across assets and applied to each asset's own feature sequence; there is no cross-asset information at decision time. With input x_t and previous state (h_{t-1}, c_{t-1}) the cell computes input, forget and output gates and a candidate cell,

$$(i_t, f_t, o_t) = \sigma\big(W_{ifo}[h_{t-1}, x_t] + b_{ifo}\big), \quad g_t = \tanh\big(W_g[h_{t-1}, x_t] + b_g\big), \tag{8}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t, \qquad h_t = o_t \odot \tanh(c_t), \tag{9}$$

with σ the logistic function and $\odot$ the Hadamard product; the gated cell state c_t is the mechanism that carries trend memory across long horizons without vanishing gradients [6]. A linear head with a tanh maps the hidden state to a position $\pi_{j,t} = \tanh\big(w^\top h_{j,t} + b\big) \in [-1, 1]$.

The network is trained by *direct reinforcement* [14]: rather than forecasting returns and converting forecasts into trades, we differentiate the realized portfolio Sharpe ratio with respect to the weights. With next-day returns ρ and cost rate c,

$$\mathcal{L} \;=\; -\frac{\bar{R}}{\mathrm{std}(R)}, \qquad R_t = \frac{1}{N}\sum_{j=1}^{N} \pi_{j,t}\, \rho_{j,t+1} \;-\; c\, \frac{1}{N}\sum_j |\pi_{j,t} - \pi_{j,t-1}|, \tag{10}$$

which is differentiable in the positions and hence, through the LSTM, in the weights. Because R_t is the return of the *whole* equal-capital portfolio, diversification enters the objective directly even though each position is decided from a single asset's features, and the turnover term $c \sum_j |\pi_{j,t} - \pi_{j,t-1}|$ discourages churn. Architecturally this is the deep momentum network of Lim et al. [10] with the changepoint augmentation of Wood et al. [16]; our contribution is the explicit Kalman/BOCPD belief block as input and the honest diversified evaluation that follows, not the network or the loss themselves. Figure 2 summarizes the architecture and the ten per-asset features.

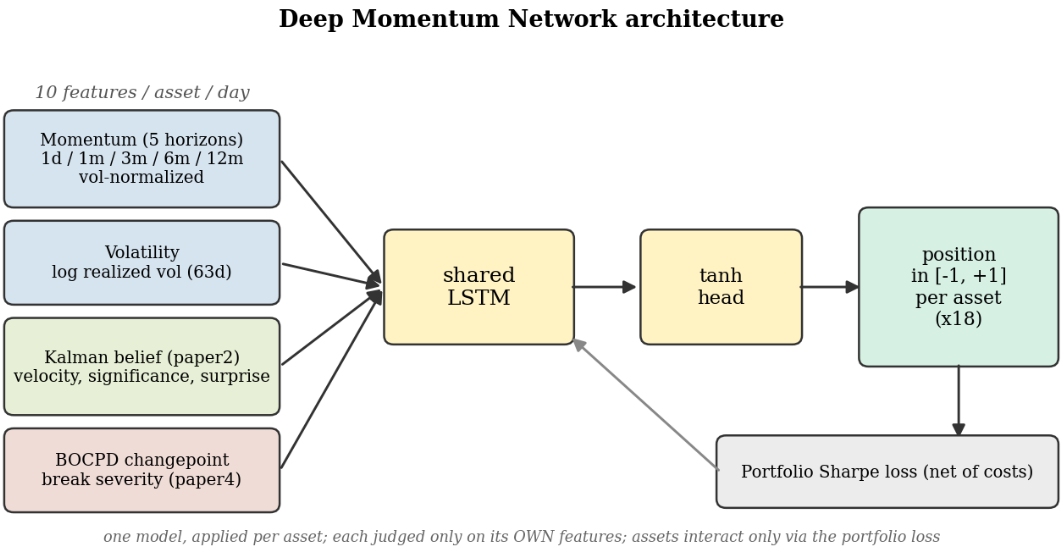

Figure 2: The belief-state Deep Momentum Network. Ten per-asset, per-day features—five volatility-normalized multi-horizon returns, log realized volatility, the three Kalman local-linear-trend belief signals, and the BOCPD changepoint severity—feed a single shared LSTM with a tanh head emitting a position in $[-1, 1]$ per asset. The network is trained on the portfolio Sharpe ratio net of turnover costs; assets interact only through that loss.

4.5 Honest nested walk-forward

We use two expanding out-of-sample folds (test blocks ending 2020 and 2024). Within each fold, the last 20% of the training window is a validation split used *both* for early stopping and for selecting the configuration (hidden size, weight decay, dropout) from a small grid; the test block is touched exactly once, for measurement. Feature standardization is fit on training data only. This makes the reported out-of-sample number free of selection bias by construction.

5 Results

Table 1 reports the headline out-of-sample comparison (2016–2024, net of costs, each return stream scaled to 10% annualized volatility for comparability). The learned Deep Momentum Network attains an information ratio of 0.58 (Newey–West $t = 1.98$; Deflated Sharpe Ratio 0.67 against five configurations), beating the fixed multi-horizon rules baseline (0.47) at a smaller maximum drawdown (-17% vs -21%). The nested procedure selected the same model size (hidden = 16) in both folds, indicating a stable rather than fragile choice.

Strategy (out-of-sample 2016–2024, net)	IR	NW-t	DSR	max DD
Rules time-series momentum (fixed)	0.47	–	–	−21%
ML LSTM Deep Momentum Network (nested)	**0.58**	1.98	0.67	**−17**%

Table 1: Honest nested out-of-sample comparison on the eighteen-ETF basket.

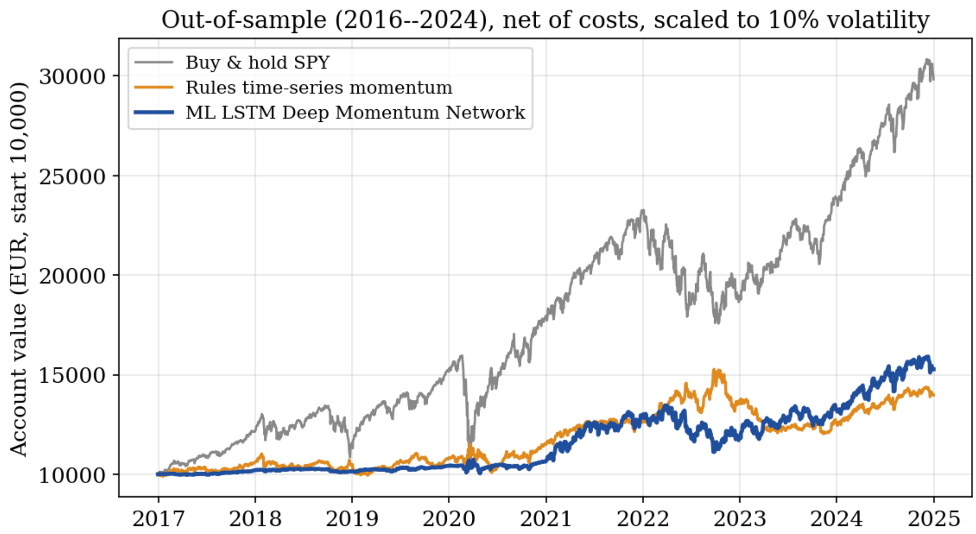

Figure 3: Out-of-sample equity curves (EUR 10,000 start, net of costs, each scaled to 10% annualized volatility). The learned Deep Momentum Network (blue) tracks above the fixed rule (orange); buy-and-hold S&P 500 (gray) is shown for reference but carries far larger drawdowns.

5.1 What does not help (and why parsimony wins)

A recurring temptation is to enrich the feature set. We tested this directly and report the result honestly: on this noisy, low-signal data, almost every addition we tried *reduced* out-of-sample performance (Table 2, single fixed configuration for a like-for-like comparison). Three findings stand out. First, volatility-normalized multi-horizon returns alone already capture most of the edge (IR = 0.55); the Kalman belief and changepoint signals raise the *statistical significance* of the result (Newey–West t from 1.56 to 2.05) more than its point estimate—they buy robustness, not raw return. Second, naive enrichments overfit: adding the Kalman signals without the changepoint, or adding a trend-acceleration feature, both dropped the out-of-sample IR sharply. Third, *technical-analysis indicators are redundant*: RSI, MACD, and Bollinger %b alone produced no out-of-sample edge (IR $= -0.02$), and appending them to the returns *hurt* ($0.55 \to 0.15$)—they are lossy transformations of the same price information the network already extracts from raw returns, consistent with the simple-beats-complex evidence of [18]. Finally, company fundamentals (value, quality, profitability) are *inapplicable* to the diversified ETF basket—an index fund has no book-to-price or ROE—and on cross-sectional equities, where they do apply, such factors are individually fragile in modern large-caps; they belong to a separate weak-signal equity study, not to this time-series setting. The consistent lesson is that the lever is the diversity of *independent assets* (Section 6), not the richness of *features*.

Feature set (single fixed config, OOS 2016–2024, net)	IR	NW-t
Momentum + volatility only (6 features)	0.55	1.56
+ Kalman belief signals (9)	0.13	0.46
+ BOCPD changepoint = full (10)	**0.56**	**2.05**
+ trend acceleration (11)	0.17	0.64
Technical indicators only (RSI/MACD/%b/vol, 4)	−0.02	−0.07
Returns + technical indicators (9)	0.15	0.61

Table 2: Feature ablation. The full belief-state set maximizes statistical significance; naive enrichments and engineered technical indicators reduce out-of-sample performance. A single fixed configuration is used throughout for comparability (nested selection could shift absolute levels; the qualitative ordering is robust).

6 Where the Edge Comes From: Diversity, Not Count

The Fundamental Law of Active Management [3] ties realized information ratio to the number of *independent* bets, not the raw instrument count. We make this concrete. A fixed time-series-momentum rule on twenty-two sector-diversified US equities returns a Sharpe ratio of only 0.15, with average pairwise correlation 0.41: across eleven sectors the names still fall together, so the effective breadth is small. Five genuinely uncorrelated diversifiers (long and intermediate Treasuries, gold, broad commodities, the dollar) have average pairwise correlation 0.03 and a standalone Sharpe of 0.19. Combining the two—twenty-seven instruments—yields a Sharpe of 0.34, *higher than either component alone*: the diversification "free lunch." Consistently, five well-chosen diversified ETFs outperform one hundred correlated equities (information ratio +0.54 vs +0.09) under the identical rule. The lesson is that asset-class diversity, measured by low cross-correlation, is the lever—not the number of names or sectors (Figure 4).

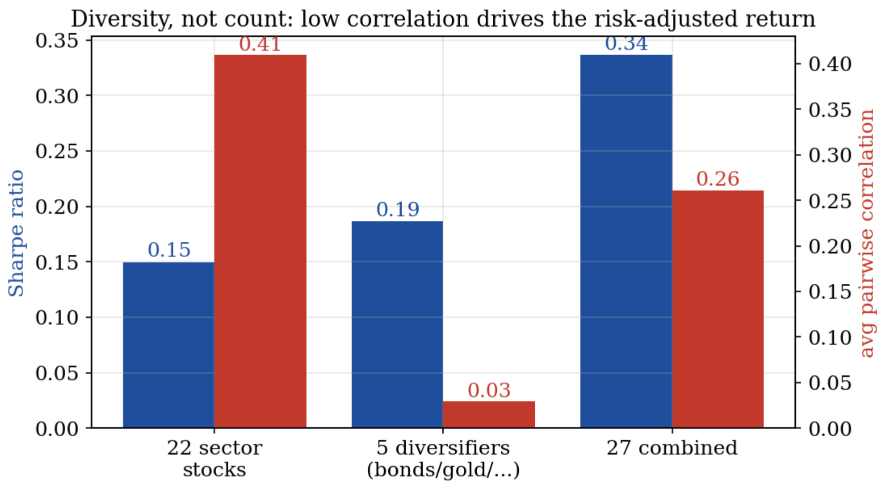

Figure 4: Sharpe ratio (blue, left axis) and average pairwise correlation (red, right axis) for twenty-two sector-diversified equities, five uncorrelated diversifiers, and their union. The combined book's Sharpe exceeds either component's: combining low-correlation streams is the "free lunch."

7 Practical Deployment: Beating Buy-and-Hold by Combining

Over 2009–2024 the US equity market was an exceptional bull; in such a regime no diversified, risk-controlled strategy beats 100% equities on *absolute* return, and ours does not in isolation.

The honest and useful question is how the trend sleeve *combines* with equities. Out of sample (2016–2024, net), a return-stacked overlay—hold the full equity position and add a half-sized trend sleeve via margin (1.5× gross)—turns EUR 10,000 into EUR 37,390 versus EUR 29,825 for buy-and-hold S&P 500, with a higher Sharpe (0.98 vs 0.84) at comparable drawdown; a leverage-free 50/50 blend has the best risk-adjusted profile (Sharpe 1.04, maximum drawdown roughly halved). The mechanism is that the trend sleeve earns precisely when equities fall (it was positive through the 2022 selloff, short bonds and equities, long the dollar and energy), so stacking it on equities adds return where it hurts most (Figures 5 and 6); we flag the obvious risk that the overlay uses leverage and that the equity–trend diversification, while historically negative in crises, is not guaranteed.

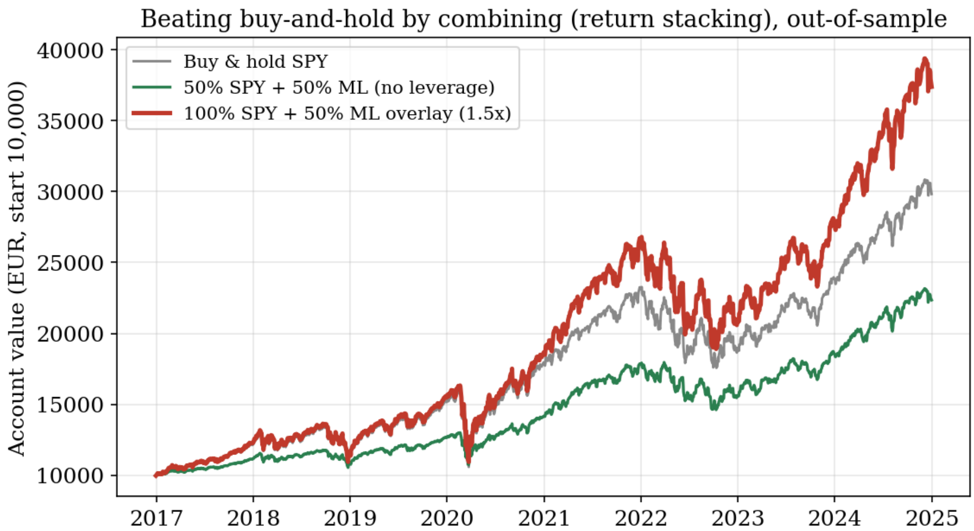

Figure 5: Out-of-sample (EUR 10,000 start, net): the return-stacked overlay (red, full equities plus a half-sized trend sleeve, 1.5× gross) ends above buy-and-hold S&P 500 (gray); the leverage-free 50/50 blend (green) is the lower-risk option.

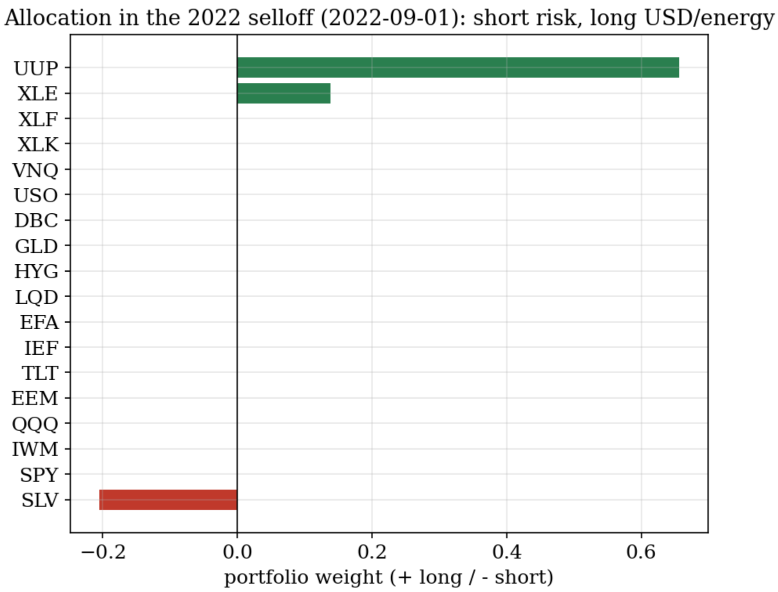

Figure 6: The strategy's positions during the 2022 selloff: short almost everything (equities, bonds, gold), long only the US dollar and energy—the two assets that rose—illustrating the long/short trend adaptation that earns when equities fall.

8 Position Sizing and Bet Optimization

Given the book, *how much to bet* is a separate question with its own literature, and it bears directly on profit. We treat it as two levers—allocation and leverage—and test both honestly out of sample (Figure 7, Table 3). For *allocation*, replacing the equal-risk inverse-volatility sizing with covariance-aware weights—minimum-variance or Hierarchical Risk Parity [11] on a Ledoit–Wolf shrunk covariance [9]— modestly improves the risk-adjusted return (Sharpe 0.44 → 0.47 → 0.52), in line with the mean–variance and risk-parity traditions [13]. For *leverage*, fractional Kelly sizing [8] (leverage $f = \text{frac} \cdot \mu/\sigma^2$, estimated on the first out-of-sample half and applied forward) raises the growth rate—quarter-Kelly ($\approx$1.8$\times$) lifts the ML book's terminal value from EUR 15,300 to EUR 20,300—but increases drawdown proportionally ($-17\% \rightarrow -30\%$). This is the honest content of the Kelly criterion: leverage scales return *and* risk together; the Sharpe ratio is invariant to leverage and improves only through better allocation or a better signal. We therefore recommend covariance-aware allocation with a *conservative* fractional Kelly—never full Kelly, which over-bets. All sizing routines are provided as a standalone module (`paper4/code/sizing.py`) usable independently of the strategy.

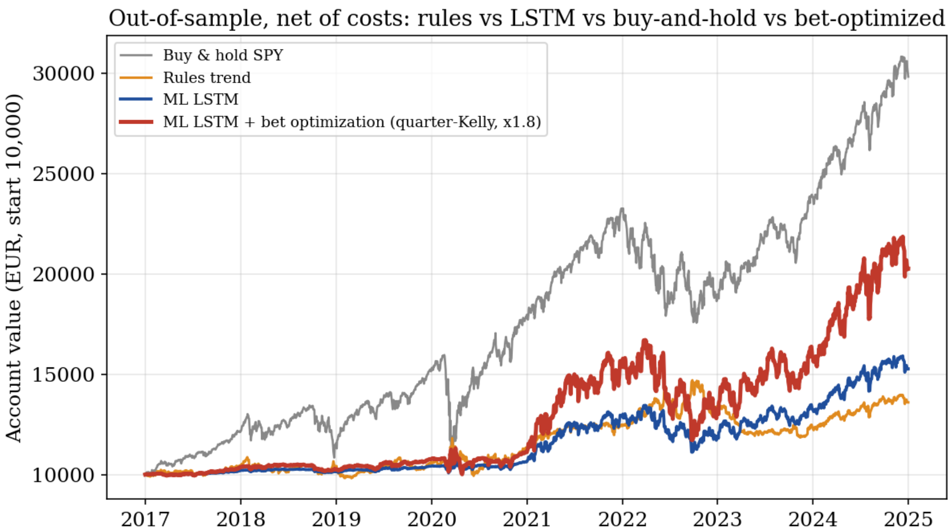

Figure 7: Out-of-sample (EUR 10,000 start, net of costs): the fixed rule, the ML LSTM, and the ML book after bet optimization (quarter-Kelly leverage, red) against buy-and-hold S&P 500. Bet sizing lifts the ML curve at the cost of larger drawdowns; it does not overtake the exceptional equity bull on raw return.

Allocation (trend signal, OOS, 10% vol target)	Sharpe	max DD
inverse-volatility (baseline)	0.44	−19%
Hierarchical Risk Parity	0.47	−28%
minimum-variance (Ledoit–Wolf)	**0.52**	−23%
Leverage (ML book)	CAGR	max DD
quarter-Kelly (×1.8)	+9.2%	−30%
half/full Kelly (×2.0 cap)	+10.1%	−33%

Table 3: Position-sizing ablation. Covariance-aware allocation improves the Sharpe ratio; fractional Kelly leverage raises return and drawdown together (Sharpe is leverage-invariant). Conservative fractional Kelly is the safe operating point.

The volatility target is the operating dial. Because the Sharpe ratio is invariant to leverage, the single deployment choice that sets the return is the annualized volatility target, which maps one-to-one to leverage. Table 4 makes the trade-off explicit on the long-only book over the real-price out-of-sample window: total return scales essentially linearly with the volatility target, and so does the maximum drawdown—there is no free lunch. We report a 10% target as the conservative default; an investor who accepts deeper drawdowns can dial it up.

Volatility target	Total return	Per year	Max drawdown
10% (conservative)	+30%	+9%	−15%
15%	+46%	+13%	−22%
20%	+63%	+18%	−28%
30% (aggressive)	+98%	+26%	−40%

Table 4: Risk-level ladder (long-only, real out-of-sample prices, net of costs). Return and drawdown both scale roughly linearly with the volatility target; the Sharpe ratio is unchanged. The target is exposed in the engine as `--target-vol` and in the backtester as `--vol`.

9 Scope and Limitations

The result is conditional on the tested universe and period; survivorship and the specific ETF set are caveats, and daily data preclude microstructure effects. The learned edge (IR ≈ 0.58, $t \approx 2$) is positive but not overwhelming, as is honest for nine years of out-of-sample diversified data. The Kalman belief is linear-Gaussian by design; genuine fat tails and stochastic volatility are only partially captured. Costs are a flat spread plus, where shorting equities, a financing charge; true execution slippage and borrow are not modeled. Consistent with prior practice, where evidence is weak we say so: the cross-sectional equity result is a null, and the time-series result is a modest, cost-surviving, but not spectacular edge.

10 Conclusion

Honest evaluation, not architecture, is the protagonist. The same momentum logic is dead on cross-sectional large-cap equities and alive on a diversified asset basket; a belief-state Deep Momentum Network, selected by a leak-free nested walk-forward, beats a strong fixed rule out of sample; the edge traces to asset-class diversity rather than instrument count; and the practical payoff is a return-stacked overlay that beats buy-and-hold equity. Every claim is reproducible from free data, net of costs, with nulls reported as nulls.

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Reproducibility. All code, the cost model, and the figures are in `etoro/paper4/` and `etoro/Strategies/slow-momentum-fast-reversion/`; results regenerate from free Yahoo data via `paper4/code/run_etf.py`.